

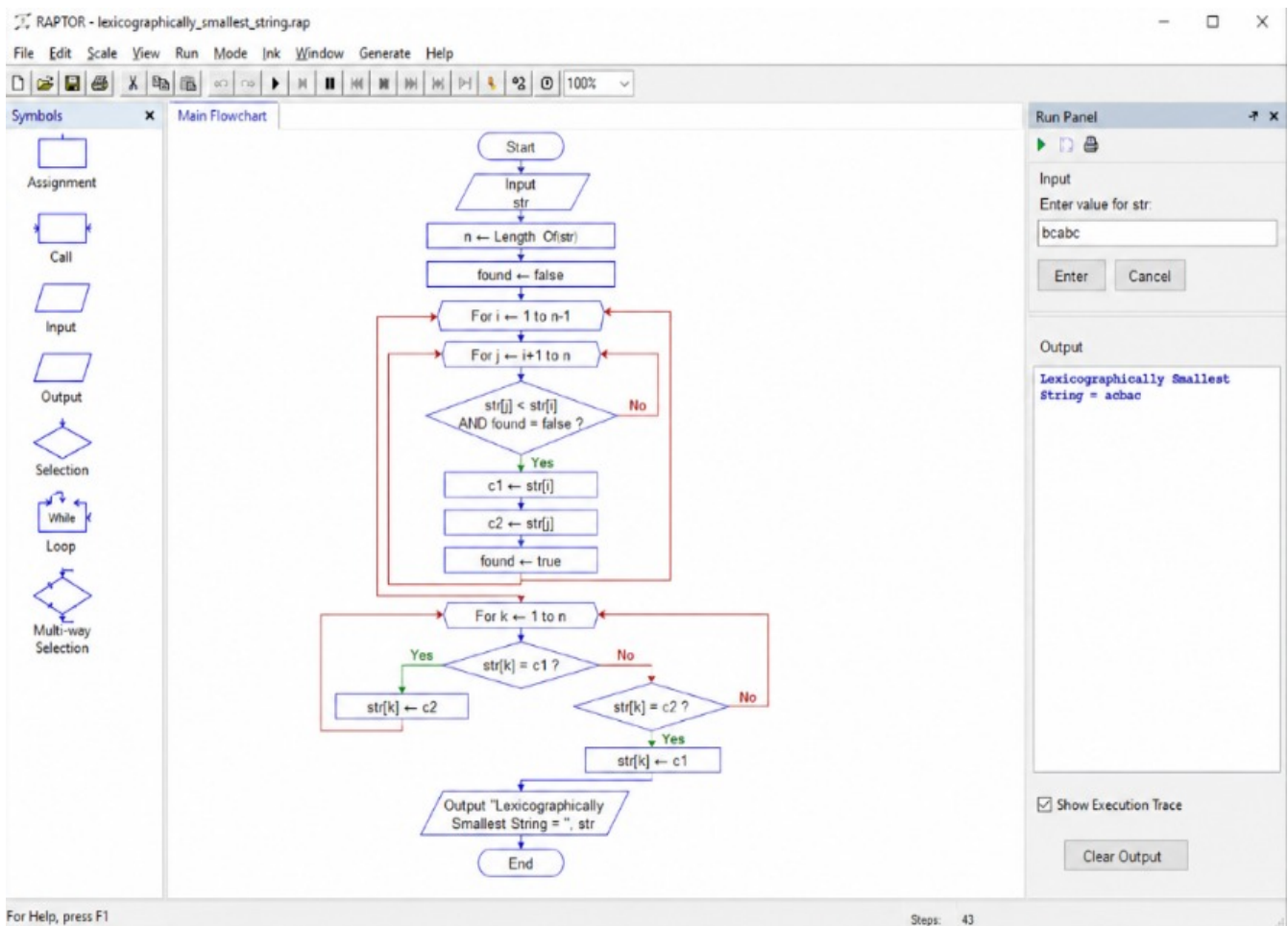
RAPTOR Flowchart for Lexicographically Smallest String

Aim: To draw and validate a RAPTOR flowchart that obtains the lexicographically smallest string by swapping all occurrences of any two characters at most once.

Theory:

A lexicographically smaller string is one that appears earlier in dictionary order. Given a string containing only lowercase English letters, one operation is allowed: choose any two characters and swap all occurrences of the first character with the second character. This operation can be performed at most once. The RAPTOR flowchart identifies a beneficial pair of characters, performs the swap throughout the string, and displays the lexicographically smallest possible string.

RAPTOR Flowchart:



Algorithm:

1. Start the program.
2. Read the input string.
3. Determine the string length.
4. Search for the first pair of characters that can produce a smaller lexicographical order.

5. Swap all occurrences of the selected two characters (only once).
6. Display the lexicographically smallest string.
7. End the program.

Sample Input: bcabc

Sample Output: Lexicographically Smallest String = abcab

Result: The RAPTOR flowchart was successfully validated and produced the lexicographically smallest string by performing the allowed character swap operation at most once.